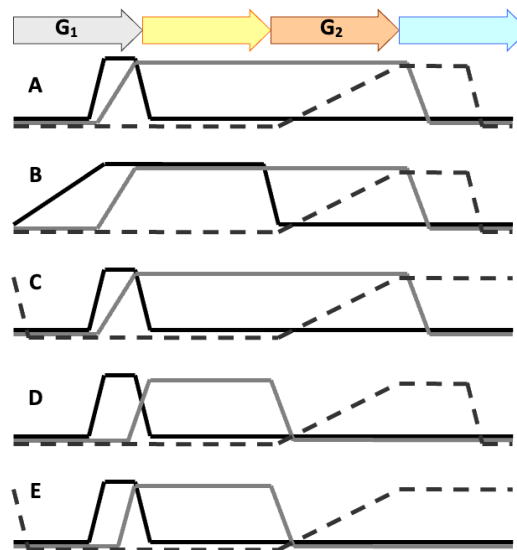


2026 Spring Cell Biology Homework (II)

- Your answers are due: **5 pm, June 4th, 2026**.
- Provide your answers on the answer sheet provided. Be sure to fill in your name and student ID number.
- The full score is 100.

I. Multiple choice questions (Select the best answer unless instructed otherwise. Questions 1~25, 3 points each; question 26, 4 points; questions 27~39, 1 point each.)

1. Which of the following simplified diagrams better shows the changes in the concentrations of three major cyclin-Cdk complexes (G1/S-Cdk, S-Cdk, and M-Cdk) in the cell in different stages of the cell cycle?



2. How many kinetochores are there in a human cell at mitosis?

- A. 23
- B. 46
- C. 69
- D. 92

3. Indicate whether each of the following descriptions better applies to a Wee1 protein (W) or a Cdc25 protein (C). Your answer would be a three-letter string composed of letters W and C only, e.g. WWC.

- () It activates M-Cdk complexes.
- () Its loss in fission yeast results in small cell size.
- () It is activated by M-Cdk complexes.

4. A cell has been subjected to ultraviolet irradiation, causing a significant number of mutations in the genome. Which of the following would you NOT expect to occur as a result?

- A. Binding of p53 to Mdm2
- B. Stabilization of p53
- C. Activation of the protein kinase Chk1
- D. Activation of the protein kinase ATR
- E. Inactivation of the protein phosphatase Cdc25

5. Which of the following events occurs in mitotic metaphase?

- A. Mitotic spindle assembly
- B. Nuclear envelope breakdown
- C. Chromosome alignment at the spindle equator
- D. Chromosome attachment to spindle microtubules for the first time
- E. Nuclear envelope reassembly

6. Which of the following mechanisms most directly contributes to signaling specificity in receptor tyrosine kinase pathways?

- A. Activation of heterotrimeric G proteins
- B. Hydrolysis of cAMP by phosphodiesterases
- C. Recruitment of distinct adaptor proteins to phosphorylated receptors
- D. Transport of steroid hormones through the bloodstream
- E. Opening of ligand-gated ion channels

7. Which of the following events is most directly triggered by IP3 production in phospholipase C signaling?

- A. Activation of Ras at the plasma membrane
- B. Phosphorylation of receptor tyrosine kinases
- C. Production of cAMP by adenylyl cyclase
- D. Release of Ca²⁺ from the endoplasmic reticulum
- E. Activation of JAK kinases

8. In most mammalian cells, low M-cyclin protein levels are maintained during G1 phase.

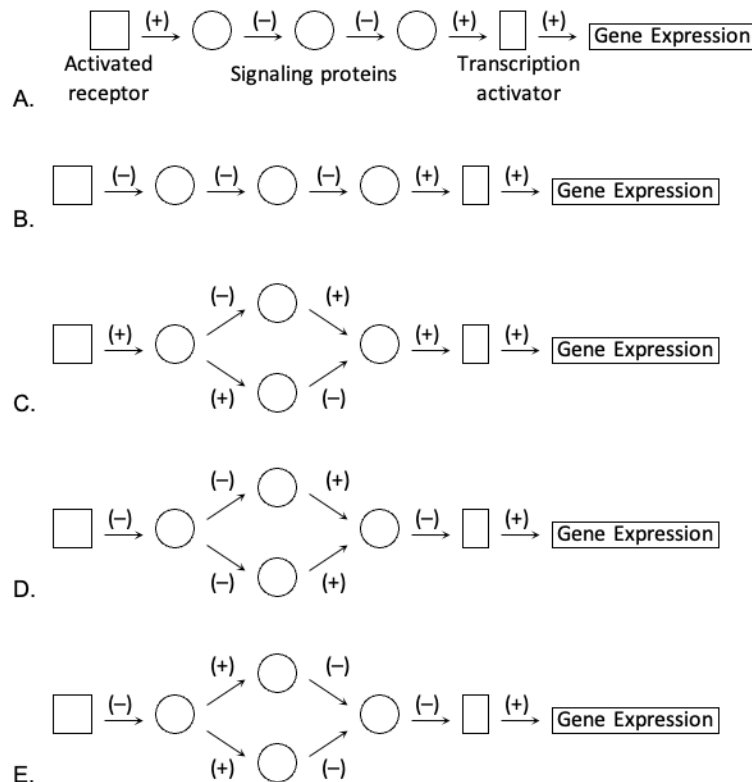
What is mainly responsible for maintaining these low levels?

- A. Cdc20–APC/C
- B. p27
- C. Cdh1–APC/C
- D. Skp2–SCF
- E. β-trCP–SCF

9. A cell expresses a transmembrane protein that is cleaved at the plasma membrane to release an extracellular fragment. The fragment binds to receptor proteins on nearby cells and activates signaling pathways resulting in altered gene expression patterns in the cells. What form of intercellular signaling does this represent?

- A. Contact-dependent signaling
- B. Paracrine signaling
- C. Synaptic signaling
- D. Endocrine signaling
- E. Autocrine signaling

10. In which of the following schematic drawings of signaling pathways does the activation of the receptor lead to gene expression? Activating and inhibitory steps are indicated with (+) and (−), respectively.



11. Consider a signaling protein that is only made up of one PH domain and two SH2 domains. This protein is most likely ...

- A. a monomeric G protein.
- B. a guanine nucleotide exchange factor.
- C. an adaptor protein.
- D. a kinase associated with receptor tyrosine kinase signaling.
- E. a negative regulator of receptor tyrosine kinase signaling.

12. Whereas the cholera toxin ADP-ribosylates the α subunit of stimulatory G protein (G_s), thereby blocking GTP hydrolysis, pertussis toxin ADP-ribosylates the α subunit of inhibitory G protein (G_i) and prevents interaction with the receptor. What is the effect of these toxins on the concentration of intracellular cAMP?

- A.They both tend to increase cAMP concentration.
- B.They both tend to decrease cAMP concentration.
- C. Cholera toxin tends to increase cAMP concentration, whereas pertussis toxin tends to decrease cAMP concentration.
- D.Cholera toxin tends to decrease cAMP concentration, whereas pertussis toxin tends to increase cAMP concentration.

13. Which of the following is a major consequence of activation of phospholipase C- β (PLC β) by the G $_q$ trimeric GTPase?

- A. Elevation of intracellular cAMP levels, leading to the activation of protein kinase A
- B. Elevation of PIP $_3$ levels in the plasma membrane, leading to the activation of protein kinase B
- C. Elevation of intracellular Ca $^{2+}$ levels, leading to the activation of protein kinase C
- D. Elevation of IP $_3$ in the plasma membrane, leading to the activation of protein kinase D
- E. Elevation of intracellular cGMP levels, leading to the activation of protein kinase G

14. What is the major way by which the monomeric G protein Ras is activated in receptor tyrosine kinase signaling?

- A. Activation of Ras-GAP
- B. Inactivation of Ras-GAP
- C. Activation of Ras-GEF
- D. Inactivation of Ras-GEF
- E. Inactivation of Ras-GDI

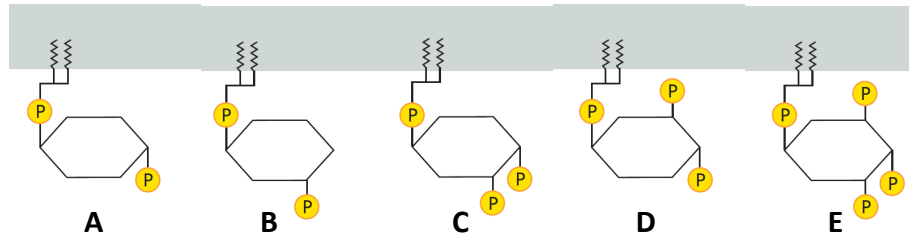
15. All nuclear receptors ...

- A. are cytosolic proteins that enter the nucleus upon ligand binding.
- B. have ligand-binding and DNA-binding domains, and can directly bind to DNA.
- C. are transcriptional activators when bound to their ligand.
- D. bind to steroid hormones.
- E. are transcriptional repressors in the absence of their ligand.

16. Sort the following events into the order that they take place in phototransduction in response to a flash of light. Your answer would be a five-letter string composed of letters A to E, e.g. BCDEA.

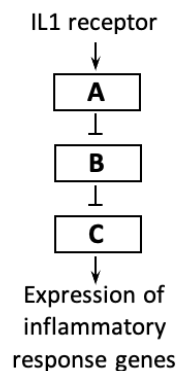
- (A) The Ca $^{2+}$ level drops.
- (B) Retinal isomerizes to an all-*trans* configuration.
- (C) G $_t$ exchanges its bound nucleotide.
- (D) The cGMP level drops.
- (E) The cGMP level rises.

17.Which of the following phosphoinositides is the preferred binding target for a PH-domain-containing protein?



18. Intracellular Ca^{2+} oscillations, initiated by the activation of IP₃ receptors and orchestrated via intricate positive and negative feedback loops, can be subject to frequency modulation, an example of which is the increase in oscillation frequency with increasing stimulus strength. Suppose compound UYA327 binds to and desensitizes the ryanodine receptors. UYA327-bound receptors can only open in response to higher calcium ion concentrations. Do you think UYA327 would tend to increase (I) or decrease (D) the frequency of calcium ion oscillations in stimulated cells? Write down I or D as your answer.

19. In the following simplified diagram of the NF κ B-dependent signaling pathway, indicate which boxes (A to C) correspond to the NF- κ B, I κ B, and IKK proteins, respectively. Your answer would be a three-letter string composed of letters A to C only, e.g. BCA.



20. During which step of the JAK-STAT signaling pathway does STAT become activated for nuclear translocation?

- A. Upon cytokine binding to a G protein-coupled receptor
- B. During receptor internalization and degradation
- C. After phosphorylation by JAK kinases following receptor dimerization
- D. Prior to ligand-induced receptor conformational change

21. All the following statements about regulation of the cell cycle are true, except:

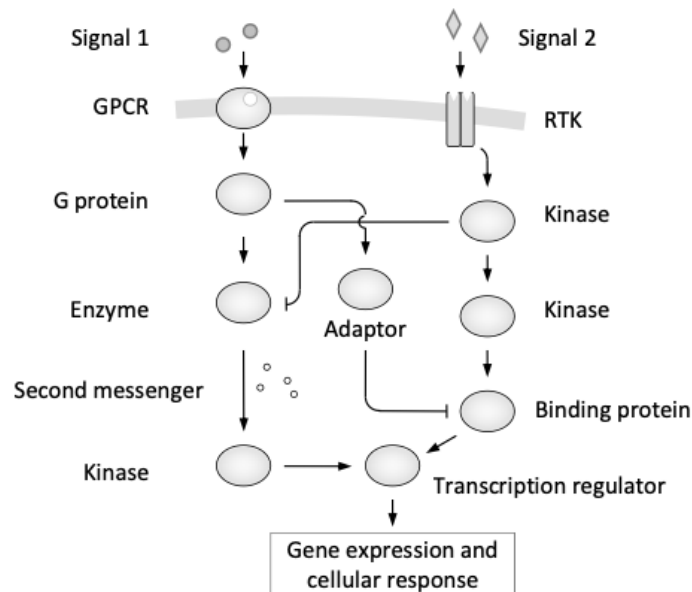
- A. Activation of cyclin-Cdk complexes requires both cyclin binding and removal of inhibitory phosphates.
- B. The anaphase-promoting complex/cyclosome (APC/C) activates Cdks by ubiquitinating cyclins.

- C. DNA damage can halt the cell cycle by stabilizing the tumor suppressor p53.
- D. Cdk activity is tightly regulated to ensure proper timing of cell cycle transitions.

22. Which of the following types of receptors typically interacts with its ligand at the cell surface?

- A. Steroid hormone receptors
- B. Toll-like receptors
- C. Nuclear hormone receptors
- D. NOD-like receptors

23. Consider the signaling pathway depicted below, which involves two different receptors. Assume that a protein is active only when it is activated but not inhibited by its upstream signaling molecules. Activation and inhibition are indicated by normal and blunt arrows, respectively. Under which of the following conditions is target-gene expression induced?



- A. Only in the presence of both signals (1 and 2)
- B. In the presence of either or both signals (1 or 2)
- C. In the presence of either signal but not both of them
- D. Only in the absence of both signals
- E. Only in the presence of signal 2 but not signal 1

24. Which of the following molecular mechanisms is NOT used in the cell in order to produce abrupt responses to a signal?

- A. Activation of a protein only when bound to multiple molecules of an activating ligand
- B. Activation of a protein only when it is phosphorylated at multiple sites
- C. Activation of a protein by simultaneously activating its activators and inactivating its inhibitors

- D. A positive feedback loop in which an activated protein positively regulates its activator
- E. A negative feedback loop in which an activated protein inhibits its activator without delay

25. Indicate true (T) and false (F) statements below regarding cytokinesis in animal cells. Your answer would be a three-letter string composed of letters T and F only, e.g. TFF.

- () The force for cytokinesis is generated by kinesin motors on microtubule bundles that form the contractile ring.
- () As the contractile ring constricts, its thickness can increase to keep a constant volume.
- () Local activation of Ran GTPase triggers the assembly and contraction of the contractile ring.

26. Indicate how each of the following motor proteins affect the length of the spindle, your answer would be a four-letter string composed of letters L (lengthen the spindle) or S (shorten the spindle) only, i.e SLSL

(1) Dynein (2) Kinesin-5 (3) Kinesin-10 (4) Kinesin-14

27. Which adhesion protein mediates adhesion between cells?

- A. Integrin
- B. Cadherin

28. The binding between cadherins for anchoring junctions is _____ dependent.

- A. Magnesium
- B. Potassium
- C. Calcium
- D. None of these

29. The epithelium can form _____ to prevent molecules from leaking across the epithelium:

- A. Gap junctions
- B. Desmosomes
- C. Adherens junctions
- D. Tight junctions

30. Cadherins interact with filaments of the cytoskeleton through an adaptor protein complex, which includes:

- A. Catenins
- B. Vinculin
- C. Actin

31. Desmosomes contain specialized cadherins that link to:

- A. Intermediate filaments
- B. Actin filaments

32. Cadherin-dependent cell sorting is an important mechanism for tissue segregation, in which the cells sort out because they express:
- A. Similar type
 - B. Different type
 - C. Complementary type
33. Gap junctions allow cells to exchange:
- A. inorganic ions
 - B. proteins
 - C. nucleic acids
34. Connexon is the basic gap-junction structure, which is composed of _____
- A. 2 subunits of connexin
 - B. 4 subunits of connexin
 - C. 6 subunits of connexin
 - D. 4 subunits of innexin
35. Glycosaminoglycans (GAGs) are composed of repeating _____ units.
- A. Disaccharide
 - B. Polysaccharide
 - C. Saccharide
36. Which of the following is not required to form proteoglycans?
- A. Glycosaminoglycans
 - B. Protein
 - C. Hyaluronan
 - D. Link tetrasaccharide
37. The collagen is made by:
- A. One collagen polypeptide chain
 - B. Two collagen polypeptide chains
 - C. Three collagen polypeptide chains
38. The basal lamina is a specialized form of extracellular matrix, which mainly contains:
- A. Fibronectin
 - B. Laminin
 - C. Elastin
 - D. Actin
39. An integrin molecule is composed of:
- A. 2 homodimeric subunits
 - B. 2 heterodimeric subunits

II. Short-answer questions

1. Epithelial-mesenchymal transition (EMT) is a physiological process in which epithelial cells acquire the motile and invasive characteristics of mesenchymal cells. Could you please briefly describe the process of EMT? (4 points)
2. Pulling on a cell-cell junction makes it stronger in one cell and will increase the contractile force generated in the attached cell. Could you please briefly describe how this tension sensing is achieved? (4 points)